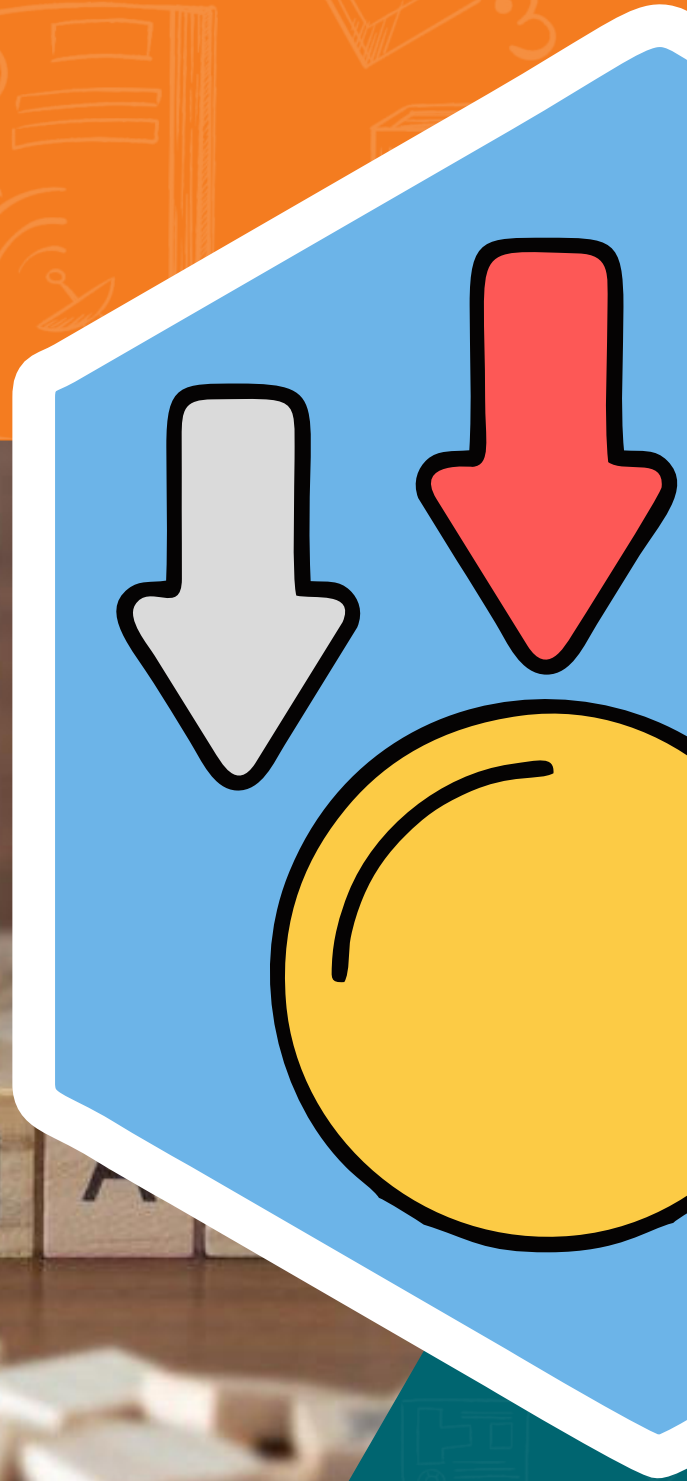


UNIT 2

KINEMATICS



SCALARS:

"All those quantities which are completely specified by only their unit and magnitude are called scalars."

Examples:

Mass, length, time, temperature, volume, density, etc.

VECTORS:

"All those quantities which are not completely specified by only their unit and magnitude as well as direction is required are called vectors."

Examples:

Force, acceleration, momentum, displacement, velocity, torque, etc.

REPRESENTATION OF VECTOR:

A vector can be represented by the following two methods:

1. Graphical representation.
2. Symbolic representation.

GRAPHICAL REPRESENTATION:

A vector quantity can be represented by an arrow-tipped line ($\longrightarrow$). In order to represent certain vector quantities, two things are essential:

1. A direction indicator.
2. A suitable scale.

SYMBOLIC REPRESENTATION:

A vector is usually represented by a letter with an arrow over it. For example, $\vec{A}$. And magnitude is represented by A or $|\vec{A}|$

TYPES OF VECTORS:**1. NEGATIVE OF A VECTOR:**

"The vector which is equal in magnitude and opposite in direction to the given vector is called the negative of a vector."

2. UNIT VECTOR:

"A unit vector in a given direction is a vector with magnitude one in that direction."

Representation:

A unit vector is represented by a letter with a cap (^) on it. For example, $\hat{a}$.

Magnitude of unit vector:

Magnitude of unit vector is unity.

Use:

Unit vector is generally used to indicate the direction of the given vector.

Formula for unit vector:

Unit vector is obtained by dividing the vector by the magnitude of the vector.

$$\hat{a} = \frac{\vec{A}}{|\vec{A}|}$$

3. FREE VECTOR (DISPLACEMENT VECTOR):

"A vector which can be displaced parallel to itself and applied at any point is known as free vector."

4. EQUAL VECTORS:

"Two vectors of the same magnitude and direction regardless of the position of their initial point are called equal vectors."

5. NULL VECTOR (ZERO VECTOR):

"A vector whose magnitude is zero and can have any arbitrary direction is called a null vector."

6. POSITION VECTOR (RADIUS OR BASE VECTOR):

"A vector which describes the position (distance and direction) of a point from the origin of the co-ordinate system is called position vector."

- Position vector is denoted by $\vec{r}$

$$\vec{r} = x\hat{i} + y\hat{j}$$

- Magnitude of position vector in two dimensions is represented by:

$$|\vec{r}| = \sqrt{x^2 + y^2}$$

ADDITION OF VECTOR:

"The process of adding two vectors in order to find the resultant vector is called the addition of vector."

RESULTANT VECTOR:

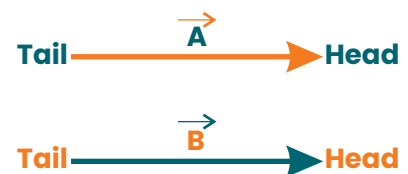
"The single vector which gives the combined effect of given vectors is called resultant vector."

GRAPHICAL METHODS OF ADDITION OF VECTOR:

1. Head to tail rule.
2. Parallelogram law.
3. Triangle law.

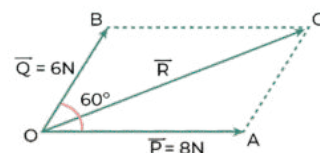
1. ADDITION OF VECTOR BY HEAD TO TAIL RULE:

- The head of the first vector is joined with the tail of the second vector.
- The head of the second vector is joined with the tail of the third vector, and so on.
- In this way, all the vectors to be added are joined together.
- The resultant is obtained by the joining of the tail of the first vector with the head of the last vector by a straight line.
- The length of this line represents the magnitude of the resultant vector.



2. PARALLELOGRAM LAW OF VECTOR ADDITION:

"If two adjacent sides of a parallelogram represent the two vectors, then the diagonal of the parallelogram will be the resultant of the two vectors."

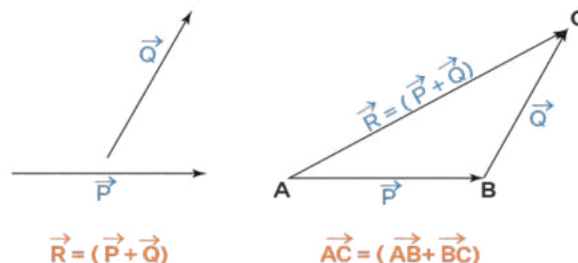


3. TRIANGLE LAW OF VECTOR ADDITION:

"The addition of two vectors by head to tail rule is called triangle law of vector addition."

LAWS/PROPERTIES FOR VECTOR ADDITION:

1. Commutative law/property.
2. Associative law/property.
3. Distributive law/property.



1. COMMUTATIVE LAW FOR VECTOR ADDITION:

"Sum of two or more vectors adding in any order will remain the same."

Proof:

Consider two vectors $\vec{A}$ and $\vec{B}$ represented by lines OA and OB. Let these two vectors represent two adjacent sides of a parallelogram. We constructed the parallelogram OACB as shown.

According to head to tail rule.

$$\vec{OA} + \vec{AC} = \vec{OC}$$

$$\vec{A} + \vec{B} = \vec{R} \text{ ----- (i)}$$

Similarly,

Comparing eq (i) and (ii) we get,

$$\vec{OB} + \vec{BC} = \vec{OC}$$

$$\vec{B} + \vec{A} = \vec{R} \text{ ----- (ii)}$$

$$\vec{A} + \vec{B} = \vec{B} + \vec{A}$$

2. ASSOCIATIVE LAW FOR VECTOR ADDITION:

"Sum of vectors adding in any group will remain the same."

Proof:

Consider three vectors $\vec{A}$, $\vec{B}$, and $\vec{C}$, represented by Op $\rightarrow$, PQ $\rightarrow$ and QS $\rightarrow$ respectively as shown in figure.

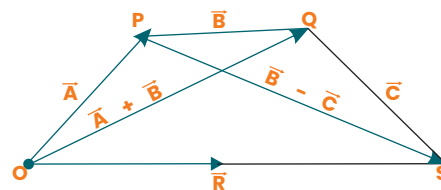
According to the head to tail rule.

$$\vec{OP} + \vec{PQ} = \vec{OQ}$$

$$\vec{A} + \vec{B} = \vec{OQ} \text{ ----- (i)}$$

$$\vec{OQ} + \vec{QS} = \vec{OS}$$

$$(\vec{A} + \vec{B}) + \vec{C} = \vec{R} \text{ ----- (ii)}$$



Similarly,

$$\vec{PQ} + \vec{QS} = \vec{PS}$$

$$\vec{B} + \vec{C} = \vec{PS} \text{ ----- (iii)}$$

$$\vec{OP} + \vec{PS} = \vec{OS}$$

$$\vec{A} + (\vec{B} + \vec{C}) = \vec{R} \text{ ----- (iv)}$$

Comparing eq (ii) and (iv) we get

$$(\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C})$$

3. DISTRIBUTIVE LAW FOR VECTOR ADDITION:

"Product of a single scalar with the resultant of two or more vectors is equal to addition of scalar with individual vectors."

$$k(\vec{A} + \vec{B}) = k\vec{A} + k\vec{B}$$

SUBTRACTION OF VECTOR:

"By subtraction of vectors we mean to find out a single vector which produces the same effect as produced by the difference of two vectors."

RESOLUTION OF VECTOR:

"The process of splitting up a single vector into two or more vectors is called resolution of vectors."

COMPONENTS:

"The vectors whose vector sum is equal to a given vector are called components of that vector."

HORIZONTAL COMPONENTS:

"The component of a vector which is directed along the x-axis is known as the horizontal component."

If F is the vector, then the horizontal component of vector F will be:

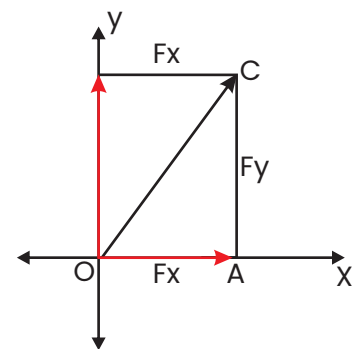
$$\cos\theta = \frac{\text{Base}}{\text{Hyp}}$$

Consider ΔOAC ,

$$\cos\theta = \frac{AO}{OC}$$

$$\cos\theta = \frac{F_x}{F}$$

$$F_x = F \cos\theta$$

**VERTICAL COMPONENTS:**

"The component of a vector which is directed along the y-axis is known as the vertical component."

If F is the vector, then the vertical component of vector F will be:

$$\sin\theta = \frac{\text{Perp}}{\text{Hyp}}$$

consider ΔOAC ,

$$\sin\theta = \frac{AC}{OC}$$

$$\sin\theta = \frac{F_y}{F}$$

$$F_y = F \sin\theta$$

COMPOSITION OF VECTOR BY RECTANGULAR COMPONENT METHOD:

If the rectangular components of a vector are given, the magnitude and direction of the vector can be obtained, thus a vector can be specified by its two rectangular components

FORMULA FOR MAGNITUDE:

$$F = \sqrt{f_x^2 + f_y^2}$$

The above formula gives us the magnitude of the resultant vector F.

FORMULA FOR DIRECTION:

$$\Theta = \tan^{-1} \frac{F_y}{F_x}$$

The above formula gives us the direction of the resultant vector F.

ADDITION OF TWO VECTORS BY RECTANGULAR COMPONENT METHOD:

Consider two vectors $\vec{V}_1$ and $\vec{V}_2$ making angles θ_1 and θ_2 with +ve x-axis respectively.

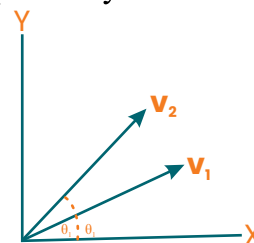
Resolve vector $\vec{V}_1$ into two rectangular components $\vec{V}_{1x}$ and $\vec{V}_{1y}$ as shown in fig:

Magnitude of these components are:

$$V_{1x} = V_1 \cos \theta_1$$

and

$$V_{1y} = V_1 \sin \theta_1$$



Resolve vector $\vec{V}_2$ into two rectangular components V_{2x} and V_{2y} as shown in figure:

Magnitude of these components are:

$$V_{2x} = V_2 \cos \theta_2$$

and

$$V_{2y} = V_2 \sin \theta_2$$

Now move vector $\vec{V}_2$ parallel to itself so that its initial point (tail) lies on the terminal point (head) of vector $\vec{V}_1$ as shown in the Fig:

Representative lines of $\vec{V}_1$ and $\vec{V}_2$ are OA and OB respectively. Join O and B which is equal to resultant vector of $\vec{V}_1$ and $\vec{V}_2$

Resultant vector along X-axis can be determined as:

$$\begin{aligned} \overline{OD} &= \overline{OC} + \overline{CD} \\ \overline{OD} &= \overline{OC} + \overline{AE} \quad \therefore \overline{CD} = \overline{AE} \\ \vec{R}_x &= \vec{V}_{1x} + \vec{V}_{2x} \end{aligned}$$

$$\vec{R}_x = V_1 \cos \theta_1 + V_2 \cos \theta_2 \dots\dots$$

Resultant vector along Y-axis can be determined as:

$$\begin{aligned} \overline{DB} &= \overline{CA} + \overline{EB} \\ \overline{OD} &= \overline{DE} + \overline{EB} \quad \therefore \overline{CA} = \overline{DE} \\ \vec{R}_y &= \vec{V}_{1y} + \vec{V}_{2y} \end{aligned}$$

$$\vec{R}_y = V_1 \sin \theta_1 + V_2 \sin \theta_2 \dots\dots$$

Now we will determine the magnitude of resultant vector by using the Pythagoras' theorem.

$$\text{Hypotenous}^2 = \text{Base}^2 + \text{Perpendicular}^2$$

$$|\overline{OB}|^2 = |\overline{OD}|^2 + |\overline{DB}|^2$$

$$R^2 = R_x^2 + R_y^2$$

$$R = \sqrt{R_x^2 + R_y^2}$$

$$R = \sqrt{(V_1 \cos \theta_1 + V_2 \cos \theta_2)^2 + (V_1 \sin \theta_1 + V_2 \sin \theta_2)^2} \dots\dots$$

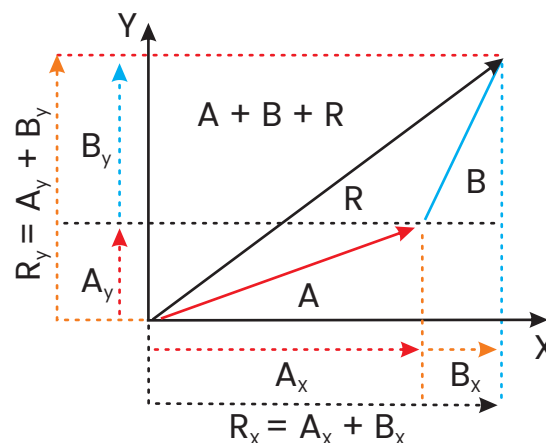
Finally the direction of resultant vector will be determined.

Again in the right angled triangle $\triangle DOB$:

$$\tan \angle DOB = \frac{\overline{DB}}{\overline{OD}}$$

$$\tan \angle DOB = \tan \theta = \frac{R_y}{R_x}$$

$$\theta = \tan^{-1} \left(\frac{R_y}{R_x} \right) \dots\dots$$



Addition of 'n' vectors by rectangular components:

Consider 'n' vectors $\vec{A}_1, \vec{A}_2, \vec{A}_3, \dots, \vec{A}_n$. All making an angle $\theta_1, \theta_2, \theta_3, \dots, \theta_n$ with the x-axis respectively.

1. Determination of x-component (A_x) of resultant:

$$\begin{aligned}\vec{A}_x &= (A_1x + A_2x + A_3x + \dots + A_nx) \hat{i} \\ A_x &= A_1x + A_2x + A_3x + \dots + A_nx \\ A_x &= A_1\cos\theta_1 + A_2\cos\theta_2 + A_3\cos\theta_3 + \dots + A_n\cos\theta_n\end{aligned}$$

2. Determination of y-component (A_y) of resultant:

$$\begin{aligned}\vec{A}_y &= (A_1y + A_2y + A_3y + \dots + A_ny) \hat{j} \\ \vec{A}_y &= A_1y + A_2y + A_3y + \dots + A_ny \\ \vec{A}_y &= A_1\sin\theta_1 + A_2\sin\theta_2 + A_3\sin\theta_3 + \dots + A_n\sin\theta_n\end{aligned}$$

3. Magnitude of resultant vector:

$$A = \sqrt{A_x^2 + A_y^2}$$

4. Direction of resultant vector:

$$\theta = \tan^{-1} \frac{F_y}{F_x}$$

PRODUCT OF VECTORS:

1. Scalar Product (Dot product).
2. Vector Product (Cross product).

SCALAR PRODUCT OR DOT PRODUCT:

"When the product of two vectors is a scalar quantity, it is called scalar product."

Representation:

The scalar product is denoted by putting a dot (.) between the two vectors.

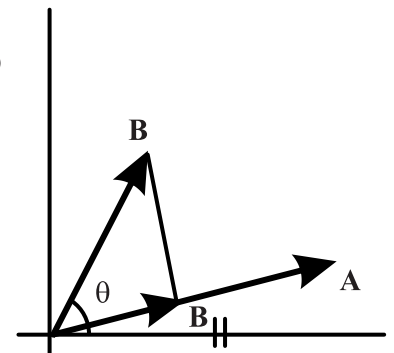
Formula:

The scalar product of two vectors A and B is given by:

$$\begin{aligned}A \cdot B &= A (\text{projection of B onto A}) \\ A \cdot B &= A \cdot B_A \quad \therefore (B_A = B\cos\theta)\end{aligned}$$

Similarly,

$$\begin{aligned}B \cdot A &= B \cdot B_A (\text{projection of A onto B}) \\ B \cdot A &= B \cdot B_A \quad \therefore (B_A = A\cos\theta) \\ A \cdot B &= |A| |B| \cos \theta\end{aligned}$$

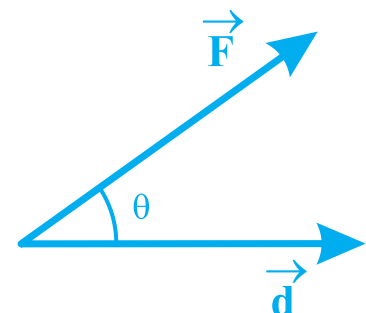


Examples of Scalars Product:

• Work:

Work is the dot product of force and displacement.

$$\begin{aligned}\text{Work} &= F \cdot d \\ W &= |F| |d| \cos \theta\end{aligned}$$



• Power:

Power is the dot product of force and velocity.

$$\text{Power} = \vec{F} \cdot \vec{V}$$

$$P = |F| |V| \cos \theta$$

• Electric potential:

Electric potential is the dot product of electric intensity and displacement.

$$\text{Electric potential} = E \cdot r$$

$$V = |E| |r| \cos \theta$$

• Electric flux:

Electric flux is the dot product of electric intensity and Area vector.

$$\text{Electric flux} = E \cdot A$$

$$\Phi_E = |E| |A| \cos \theta$$

• Magnetic flux:

Magnetic flux is the dot product of magnetic field induced and Area vector.

$$\text{Magnetic flux} = B \cdot A$$

$$\Phi_B = |B| |A| \cos \theta$$

Characteristics of Dot Product:

• Maximum or Positive scalar product:

Scalar product of two vectors A and B is maximum when the angle between the two non-zero vectors is zero or vectors are parallel or vectors are in same direction.

$$A \cdot B = |A| |B|$$

Minimum or zero scalar product:

Scalar product of two vectors A and B is minimum when the angle between the two non-zero vectors is 90° or vectors are perpendicular.

$$A \cdot B = 0$$

Negative scalar product:

Scalar product of two vectors A and B is negative when the angle between the two non-zero vectors is 180° or vectors are in opposite direction.

$$A \cdot B = -|A| |B|$$

Dot product of two non-zero perpendicular (unit) vectors:

The scalar product of two perpendicular vectors is always zero.

$$\hat{i} \cdot \hat{i} = 1 \quad \hat{k} \cdot \hat{k} = 1 \quad \hat{j} \cdot \hat{j} = 1$$

Dot product of two equal and parallel (unit) vectors:

If two vectors are equal and parallel, then the angle between them is zero.

$$\hat{i} \cdot \hat{i} = 1 \quad \hat{k} \cdot \hat{k} = 1 \quad \hat{j} \cdot \hat{j} = 1$$

Non-zero equal vectors which are parallel (But not unit vectors) then:

$$A \cdot A = |A|^2$$

Dot product of two unequal non-zero vectors (In terms of rectangular components):

Consider two vectors A and B in terms of rectangular components:

$$A = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$B = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

Their dot product is given by:

$$A \cdot B = B \cdot A$$

Commutative law for dot product:

According to the commutative law for dot product:

$$A \cdot B = B \cdot A$$

Distributive law for dot product:

According to the distributive law for dot product:

$$A \cdot (B + C) = A \cdot B + A \cdot C$$

VECTOR PRODUCT OR CROSS PRODUCT:

"When the product of two vectors is also a vector quantity, it is called vector product."

OR

"Vector product or cross product of two vectors is equal to the product of the magnitude of one vector A with the component of the second vector perpendicular to the first vector."

Representation:

The vector product is denoted by putting a cross (X) between the two vectors.

Formula:

The vector product of two vectors A and B is given by:

Where u is unit vector pointing in the direction of the product vectors.

$$A \times B = |A| |B| \sin \theta_{AB} \hat{u}$$

Formula in terms of rectangular components:

The given vectors A and B in terms of rectangular components can be written as:

$$A = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$B = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

Then,

$$A \times B = \begin{vmatrix} i & j & k \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

Magnitude of vector product:

The magnitude of vector product of two vectors A and B is given by:

$$|A \times B| = |A| |B| \sin \theta_{AB}$$

$$|A \times B| = |A| |B| \sin \theta_{AB}$$

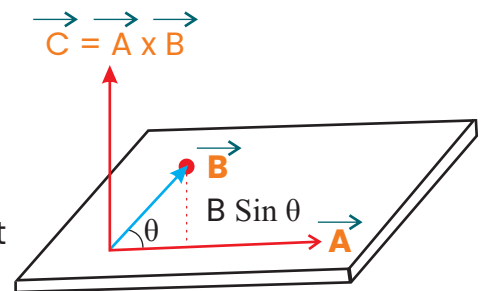
Examples of vector product:

• Torque:

Torque is the cross product of force and force arm.

$$\text{Torque} = r \times F$$

$$\tau = |r| |F| \sin \theta$$



• Angular momentum:

Angular momentum is the cross product of position vector and linear momentum.

$$\text{Angular momentum} = \vec{r} \times \vec{p}$$

$$L = |r| |p| \sin\theta$$

Force on a charge particle in a magnetic field:

An electric charge 'q' moving with velocity v in a magnetic field B experiences a force F which is given by:

$$\text{Force} = q (\vec{V} \times \vec{B})$$

$$F = q |V| |B| \sin(\theta)$$

Characteristics of cross product:

• Maximum vector product:

Vector product of two vectors A & B is maximum when the angle between the two non-zero vectors is 90° or vectors are perpendicular.

$$A \times B = |A| |B| \hat{u}$$

• Minimum vector product:

Vector product of two vectors A & B is minimum when the angle between the two non-zero vectors is 0° or vectors are parallel.

$$A \times B = 0$$

• Cross product of two non-zero parallel vectors:

The cross product of two parallel vectors is always zero.

Example:

Three unit vectors $\hat{i}, \hat{j}, \hat{k}$ are mutually perpendicular to each other then:

$$\hat{i} \times \hat{i} = 0 \quad \hat{j} \times \hat{j} = 0 \quad \hat{k} \times \hat{k} = 0$$

Example:

$$\vec{A} \times \vec{A} = 0$$

• Cross product of perpendicular unit vectors:

Three unit vectors $\hat{i}, \hat{j}, \hat{k}$ are mutually perpendicular to each other then:

$$\hat{j} \times \hat{i} = -\hat{k} \quad \hat{k} \times \hat{j} = -\hat{i} \quad \hat{i} \times \hat{k} = -\hat{j}$$

• Commutative law for cross product (Cross product is not commutative):

Consider two vectors A and B making an angle θ with each other,

$$\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A}) \quad \text{OR} \quad \vec{A} \times \vec{B} \neq \vec{B} \times \vec{A}$$

DISPLACEMENT:

"Change of position of a body in a particular direction is called displacement."

Representation:

It is denoted by 'd'.

Nature:

- It is a vector quantity.
- Displacement can be negative. It contains direction.

Unit:

In S.I system its unit is meter.

Dimension:

Its dimension is L.

SPEED:

"Distance covered in unit time is called speed."

Representation:

It is denoted by 'V'.

Formula:

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

$$\text{Speed} = \frac{s}{t}$$

Nature:

- It is a scalar quantity.

Unit:

In S.I system its unit is meter per second (ms^{-1})

Dimension:

Its dimension is L T^{-1}

TYPES OF SPEED:**1. Instantaneous speed:**

"Speed of a body at a particular instant is called instantaneous speed."

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t}$$

2. Uniform speed:

"If a body covers equal distances in equal intervals of time then its speed is said to be uniform."

- If a body is moving with uniform velocity, then $V_{av} = V_{inst}$

3. Variable speed:

"If a body covers unequal distances in equal intervals of time then its speed is said to be variable."

- If a body is moving with non-uniform velocity, then $V_{av} \neq V_{inst}$

4. Average speed:

"Average distance covered by a body in unit time is called average speed."

$$\text{Average Speed} = \frac{\text{total distance}}{\text{total time}}$$

VELOCITY:

"Change of displacement of a body with respect to time is called velocity."

Representation:

It is denoted by $\vec{V}$.

Formula:

$$\text{Velocity} = \frac{\text{displacement}}{\text{time}}$$

$$v = \frac{\Delta r}{\Delta t}$$

Nature:

It is a vector quantity.

Direction:

Direction of velocity is along the direction of displacement.

Unit:

In S.I system its unit is meter per second (ms^{-1})

Dimension:

Its dimension is L T^{-1}

Displacement time graph:

When we plot the displacement of a moving body from some fixed point against time (t), a displacement time graph of the motion of the body is obtained.

Calculation of velocity from displacement time graph:

The velocity of a body can be determined from the slope of the displacement time graph.

Determination of uniform velocity:

When a body moves with uniform velocity, it covers equal displacement in equal intervals of time. A graph of displacement against time will be a straight line as shown in the figure. If we take a point on the graph and draw a perpendicular AB on the time axis, the line AB will represent the displacement and OB will give the time during which the body covered the displacement AB. Hence, the velocity of the body is given by:

$$\text{Velocity} = \frac{\text{displacement}}{\text{time}} = \frac{AB}{OB}$$

Determination of variable velocity:

If the body has variable velocity, its velocity at any instant 't' is found by drawing a tangent EG at the point A at that instant and obtaining a right-angled triangle EFG. The slope of the line at A will give the velocity of the body.

$$\text{Velocity at A} = \frac{GF}{EF}$$

TYPES OF VELOCITY:**1. Average velocity:**

"Ratio of total displacement to the time taken to cover this displacement is called average velocity."

2. Instantaneous velocity:

"The velocity at any instant of time is called instantaneous velocity."

3. Uniform velocity:

"If a body covers equal displacement in equal intervals of time is called uniform velocity."

4. Variable velocity:

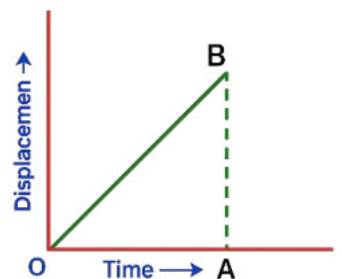
"If a body covers equal displacement in equal intervals of time is called uniform velocity."

ACCELERATION:

"Change of velocity in unit time is called acceleration."

Representation:

It is denoted by $\vec{a}$.



Displacement-time graph for an object moving with uniform velocity

Formula:

$$\text{Acceleration} = \frac{\text{Change in velocity}}{\text{Unit time}}$$

$$\text{Acceleration} = \frac{\Delta \vec{v}}{\Delta t}$$

Nature:

It is a vector quantity.

Direction:

Direction depends on motion.

Unit:

In S.I system its unit is meter per second (ms^{-2})

Dimension:

Its dimension is L T^{-2}

Velocity time graph:

The acceleration of a body can be determined from the slope of the velocity time graph.

Determination of acceleration from velocity time graph:

Acceleration of the body at a given point on its velocity time graph is equal to the slope of the tangent to the graph at that point.

Acceleration of the body at point 'P' is equal to the slope of tangent AB to the curve, at 'P' the acceleration is:

TYPES OF ACCELERATION:**1. Average acceleration:**

"The rate of change of velocity over any interval of time is called its average acceleration."

$$a_{av} = \frac{v_2 - v_1}{t_2 - t_1}$$

2. Instantaneous acceleration:

"The acceleration of a body at particular instant or during very short interval of time is called instantaneous acceleration."

$$a_{inst} = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t}$$

3. Uniform acceleration:

"If a velocity of body changes equally in equal interval of time then the acceleration is called uniform acceleration."

4. Variable acceleration:

"If the velocity of the body changes unequally in equal interval of time then the acceleration is said to be variable acceleration."

5. Positive acceleration:

"If the velocity of the body increases then the acceleration produced in a body is called positive acceleration."

6. Negative acceleration OR Retardation OR Deceleration:

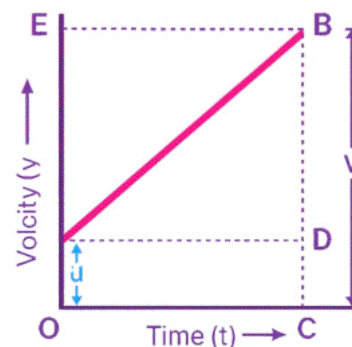
"If the velocity of the body decreases then the acceleration produced in a body is called negative acceleration."

Derivation of First Equation of Motion by Graphical Method

The first equation of motion can be derived using a velocity-time graph for a moving object with an initial velocity of u , final velocity v , and acceleration a .

In the graph:

- The velocity of the body changes from A to B in time t at a uniform rate.
- BC is the final velocity and OC is the total time t .
- A perpendicular is drawn from B to OC, a parallel line is drawn from A to D, and another perpendicular is drawn from B to OE (represented by dotted lines).



The following details are obtained from the graph above:

- The initial velocity of the body, $u = OA$
- The final velocity of the body, $v = BC$

From the graph, we know that

$$BC = BD + DC$$

Therefore,

$$\begin{aligned} v &= BD + DC \\ v &= BD + OA \quad (\text{since } DC = OA) \end{aligned}$$

Finally,

$$v = BD + u \quad (\text{since } OA = u) \quad (\text{Equation 1})$$

Now, since the slope of a velocity-time graph is equal to acceleration a .
So,

$$\begin{aligned} a &= \text{slope of line AB} \\ a &= BD/AD \end{aligned}$$

Since $AD = OC = t$, the above equation becomes:

$$BD = at \quad (\text{Equation 2})$$

Now, combining Equation 1 & 2, the following is obtained:

$$v = u + at$$

Derivation of Second Equation of Motion by Graphical Method:

From the graph above, we can say that

Distance travelled (s) = Area of figure OABC = Area of rectangle OADC + Area of triangle ABD

$$S = (1/2 \times AD \times BD) + (OA \times OC)$$

As $OA = u$ and $OC = t$, the above equation becomes,

$$S = (1/2 \times AD \times BD) + (u \times t)$$

As $BD = at$ (from the graphical derivation of 1st equation of motion), the equation becomes,

$$S = (1/2 \times t \times at) + ut$$

On further simplification, the equation becomes:

$$s = ut + (1/2) at^2$$

Derivation of Third Equation of Motion by Graphical Method:

From the graph, we can say that

The total distance travelled; S is given by the Area of trapezium OABC.

Hence,

$$S = (1/2) \times (\text{Sum of Parallel Sides}) \times \text{Height}$$

$$S = (1/2) \times (OA + CB) \times OC$$

Since $OA = u$, $CB = v$, and $OC = t$, the above equation becomes:

$$S = (1/2) \times (u + v) \times t$$

Now, since $t = (v - u) / a$,

The above equation can be written as:

$$S = (1/2) \times ((v - u) / a) \times (v + u)$$

$$S = (1/2) \times ((v^2 - u^2) / a)$$

$$S = (v^2 - u^2) / (2a)$$

Now, multiplying both sides by $2a$:

$$2aS = v^2 - u^2$$

The third equation of motion is obtained by solving the above equation:

$$v^2 = u^2 + 2aS$$

PROJECTILE:

"Any object thrown at some angle θ above or below the horizontal and moves free under the action of gravity is called projectile."

PROJECTILE MOTION:

"If a body moves such that it has constant velocity in horizontal direction and at the same time it falls freely under the action of gravity then its motion is known as projectile motion."

Examples:

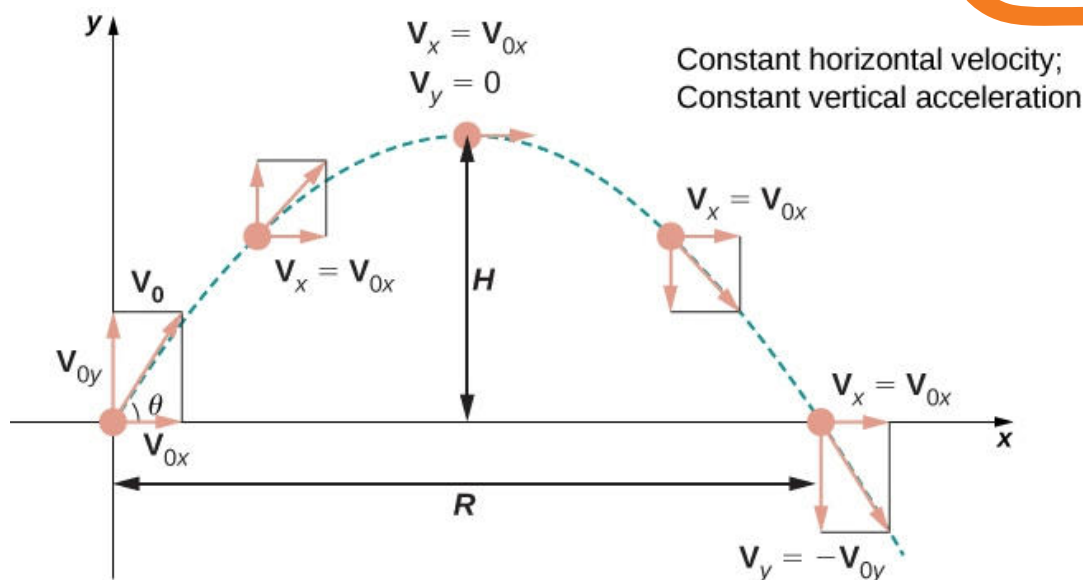
- A bullet fired from a gun.
- A football kicked in the air.
- Jumping animal.
- Object thrown from a window.
- A bomb dropped from an air plane.

Assumptions for projectile motion:

1. Acceleration considers only in vertical path; horizontal motions are uniform.
2. The acceleration due to gravity 'g' is constant throughout the flight and is directed downward.
3. Friction and other than gravitational forces are negligible.
4. The rotation of earth does not affect projectile motion.
5. The earth is considered as flat.

MATHEMATICAL ANALYSIS OF PROJECTILE MOTION:

Consider a projectile through in gravitational field of earth with velocity V_0 at an angle θ with horizontal axis. At any instant the velocity of the projectile has two components.



Horizontal component	Vertical component
VELOCITY	VELOCITY
$V_{0x} = V_0 \cos \theta$	$V_{0y} = V_0 \sin \theta$
$V_x = V_{0x}$	$V_y = V \sin \theta$
ACCELERATION	ACCELERATION
$a_x = 0$	$a_y = -g$
DISPLACEMENT	DISPLACEMENT
$x = V_{0x} t$	$y = V_{0y} t - \frac{1}{2} g t^2$

PARAMETERS OF PROJECTILE:

1. TIME TO REACH THE MAXIMUM HEIGHT (t):

Definition:

"Time taken by projectile to reach the maximum height is called time to reach the maximum height."

Derivation:

Time to reach maximum height can be calculated by considering the vertical motion of the projectile.

- Initial vertical velocity = $V_{0y} = V_0 \sin \theta$
- Final velocity (at max. height) = 0
- Acceleration along y-axis = $-g$

$$\begin{aligned}
 V_f &= V_i + at \\
 0 &= V_0 \sin \theta - gt \\
 gt &= V_0 \sin \theta \\
 \mathbf{t} &= \mathbf{V_0 \sin \theta / g}
 \end{aligned}$$

2. TOTAL TIME OF FLIGHT (T):

Definition:

"Total time taken by projectile from its point of projection to the point where it strikes the ground is called total time of projectile."

Derivation:

It can be calculated by considering the vertical component of motion.

- Initial vertical velocity = $V_{0y} = V_0 \sin \theta$
- Acceleration along y-axis = $-g$
- Distance (Vertical) = $h = 0$
- Total time of flight = T

Displacement (Y) of the projectile at any time t along Y-axis is given by:

$$Y = V_{0y}t + \frac{1}{2}at^2$$

$$0 = V_0 \sin \theta T - \frac{1}{2}gT^2$$

$$\frac{1}{2}gT^2 = V_0 \sin \theta T$$

$$T = \frac{2V_0 \sin \theta}{g}$$

3. MAXIMUM HEIGHT OF PROJECTILE:

Definition:

"Maximum vertical distance covered by the projectile is called maximum height of projectile."

Derivation:

- Initial vertical velocity $V_{0y} = V_0 \sin \theta$
- Acceleration along y-axis = $a = -g$
- Distance (Vertical) = h

Time to reach maximum height = $T = V_0 \sin \theta / g$

- The displacement (Y) of the projectile at any time t along Y-axis is given by:

$$Y = V_{0y}t + \frac{1}{2}at^2$$

$$h = (V_0 \sin \theta) (V_0 \sin \theta / g) - \frac{1}{2}g (V_0 \sin \theta / g)^2$$

$$h = V_0^2 / g \sin^2 \theta - V_0^2 \sin^2 \theta / 2g$$

$$h = (2V_0^2 \sin^2 \theta - V_0^2 \sin^2 \theta) / 2g$$

$$h = (V_0^2 \sin^2 \theta) / 2g$$

4. RANGE OF PROJECTILE:

Definition:

"Distance travelled by the projectile along the horizontal direction from the point of projection is called range of projectile."

Derivation:

It can be calculated by considering the horizontal component of motion of the projectile.

- Horizontal velocity $V_{0x} = V_0 \cos \theta$
- Total time of flight = $T = 2V_0 \sin \theta / g$
- Horizontal range = $x = R$

$$S = V \times T$$

$$R = V_{0x} \times T$$

$$R = (V_0 \cos \theta) (2V_0 \sin \theta / g)$$

$$R = (V_0^2 / g) 2 \sin \theta \cos \theta$$

$$\therefore 2 \sin \theta \cos \theta = \sin 2\theta$$

$$R = (V_0^2 / g) \sin 2\theta$$

5. MAXIMUM HORIZONTAL RANGE (R_{max}):

Definition:

"Maximum distance travelled by the projectile along the horizontal direction from the point of projection is called maximum horizontal range."

Derivation:

The horizontal range (R) of projectile depends upon the projection angle (θ) and initial speed V_0 . For a given initial speed V_0 , the horizontal range will be maximum when,

$$\begin{aligned}\sin 2\theta &= 1 \\ 2\theta &= \sin^{-1}(1) \\ \theta &= 90^\circ\end{aligned}$$

Therefore, the maximum horizontal range,

$$\begin{aligned}R_{\max} &= (V_0^2 / g) \times \sin 2\theta \\ R_{\max} &= (V_0^2 / g) \times \sin 2(45^\circ) \\ R_{\max} &= (V_0^2 / g) \times \sin(90^\circ) \\ \therefore \sin(90^\circ) &= 1 \\ \mathbf{R_{\max} = (V_0^2 / g)}\end{aligned}$$

APPLICATIONS OF PROJECTILE MOTION:**1. Projectile in athletics:**

- The projectile motion formula can be used to predict best serving angle.
- Many athletics games such as baseball, football, hockey, cricket etc. involving projectile motion that thrown, kicked or struck can be discussed using projectile motion formula.

2. Horizontal motion:

- Using the value of initial velocity of jumper and the angle at which the jumper projected himself, we can evaluate ranges.

3. Flat and high trajectory:

- We can use small and large angles for flat and high trajectory by using projectile formulas.